Reporting module refactoring

EXERCISE 3 | PHASE 4 | JAVA 17 / MAVEN | 7 OCTOBER 2026

48 tests passed	**13 to 0** unsuppressed PMD findings	**1** page summary

Original problems

Fifteen mapped findings: mixed responsibilities (SRP), format branching (OCP), concrete side effects (DIP), missing small interfaces (ISP), duplicated delivery (DRY), pointless delay (KISS), swallowed exceptions, growing string copies, broken empty JSON, missing JSON/CSV escaping, ambiguous null returns, weak naming, poor logging and difficult isolated testing.

Redesigned architecture

ReportApplication -> ReportGenerator -> ReportFormatter + ReportSender. JSON/CSV implement one formatter contract. MonitoringReportSender wraps an injected synchronous transport. Constructor injection separates coordination, serialization and delivery. Both interfaces have one operation.

Refactored code

Seven production files contain the two interfaces, two formatters, sender, generator and CLI. Six test/support files contain five JUnit suites and log capture. Core excerpt from **ReportGenerator.generateReport**; surrounding validation, logging and stage tracking are omitted. Full implementation: **src/main/java**.

```
String report = Objects.requireNonNull(formatter.format(data), "formatter returned null");
sender.send(report);
return report;
```

Craftsmanship improvements

Cohesive methods, final dependencies, StringBuilder/StringJoiner, explicit contracts and JDK event logging replace duplication and diagnostic prints. Ordinary JSON/CSV output, order and value-to-string conversion are preserved. Escaping, empty JSON, fail-fast validation and removal of the delay are intentional fixes. Original exceptions propagate without hidden retries.

Actual verification and static analysis

OpenJDK 17.0.19 / Maven 3.9.10: compilation, executable JAR and mvn verify succeeded; **48 tests passed**, with 0 failures, errors or skips. **PMD 7.17.0 / plugin 3.28.0, 20 focused rules:** original 13 findings (10 console calls, 2 string-append warnings, 1 empty catch); refactored 0 unsuppressed findings, 1 documented CLI-output suppression, 0 analysis errors. Two initial rule-reference errors were corrected; genuine logs/XML are saved in verification. JAR demos, five legacy-output comparisons and independent JSON/CSV parser round trips passed. Monitoring remains an explicitly labeled local simulation.

Integration benefits

Formats and transports evolve independently; fakes isolate failure paths and stable event/stage fields expose delivery failures. One shared workflow reduces inconsistent changes. Committed tests and analysis rules support maintenance. Detailed analysis, architecture, review and evidence accompany the code.

Review scope: the separate referenced checklist was absent. This automated implementation/review does not replace the assignment's required student team work.